

Exercise on Hypothesis Testing

The purpose of this exercise is to design a statistical test to discover a signal process by counting events in a detector. The exercise involves calculations by hand as well as some computations using python. The program `hypTest.py` can be used as a starting point for parts (a)–(f) and `hypTestMC.py` can be used for part (g).

Suppose a detector that looks, e.g., for Dark Matter interactions can for each event measure a quantity x with $0 \leq x \leq 1$. The events can be of two types: signal (s) or background (b). The probability density functions for the s and b events are

$$f(x|s) = 3(1-x)^2, \quad (1)$$

$$f(x|b) = 3x^2. \quad (2)$$

1(a) Suppose for each event we test the hypothesis that it is background. We reject this hypothesis if the observed value of x is less than a specified cut value x_{cut} . Find the value of x_{cut} such that the probability $P(x < x_{\text{cut}}|b)$ to reject the background hypothesis (i.e., accept as signal) if it is background is $\alpha = 0.05$. (The value α is the *size* or significance level of the test used to select events.)

1(b) For the value of x_{cut} that you find, what is the probability $P(x < x_{\text{cut}}|s)$ to reject the background hypothesis (i.e., accept as a candidate signal event) with $x < x_{\text{cut}}$ given that it is signal. (This is the *power* of the test of the background hypothesis with respect to the signal alternative or equivalently the signal efficiency.)

1(c) Suppose that the expected number of background events is $b_{\text{tot}} = 100$ and for a given signal model one expects $s_{\text{tot}} = 10$ signal events. Find the expected numbers of events s and b of signal and background events that will satisfy $x < x_{\text{cut}}$ using the value of $x_{\text{cut}} = 0.1$, i.e.,

$$s = s_{\text{tot}}P(x < x_{\text{cut}}|s), \quad (3)$$

$$b = b_{\text{tot}}P(x < x_{\text{cut}}|b). \quad (4)$$

1(d) Assuming the numbers from 1(c), the prior probabilities for an event to be signal or background are

$$\pi_s = \frac{s_{\text{tot}}}{s_{\text{tot}} + b_{\text{tot}}} = 0.09, \quad (5)$$

$$\pi_b = \frac{b_{\text{tot}}}{s_{\text{tot}} + b_{\text{tot}}} = 0.91. \quad (6)$$

Using Bayes' theorem with these values, find the probability for an event to be signal given that it has $x < x_{\text{cut}}$ (the signal purity of the selected sample).

1(e) Suppose for a certain x_{cut} one has $b = 0.5$ and we find there $n_{\text{obs}} = 3$ events in the search region $x < x_{\text{cut}}$. We want to test the hypothesis that $s = 0$ (the background-only hypothesis or “b”), against the alternative that signal is present with $s \neq 0$ (the “ $s + b$ ” hypothesis).

The actual number of events n found in the experiment with $x < x_{\text{cut}}$ can be modeled as following a Poisson distribution with a mean value of $s + b$. That is, the probability to find n events is

$$P(n|s, b) = \frac{(s + b)^n}{n!} e^{-(s+b)} . \quad (7)$$

The p -value of the background-only hypothesis is the probability, assuming $s = 0$, to find $n \geq n_{\text{obs}}$:

$$p = P(n \geq n_{\text{obs}} | s = 0, b) = \sum_{n=n_{\text{obs}}}^{\infty} \frac{b^n}{n!} e^{-b} = 1 - \sum_{n=0}^{n_{\text{obs}}-1} \frac{b^n}{n!} e^{-b} . \quad (8)$$

Find the p -value using the values given above and from this find the *significance* with which one can reject the $s = 0$ hypothesis, defined as

$$Z = \Phi^{-1}(1 - p) , \quad (9)$$

where Φ is the standard cumulative Gaussian distribution and Φ^{-1} is its inverse (the standard Gaussian quantile).

1(f) The expected (median) significance assuming the $s + b$ hypothesis of the test of the $s = 0$ hypothesis is a measure of sensitivity and this is what one tries to maximize when designing an experiment. It can be approximated with a number of different formulas. For $s \ll b$ one can use $\text{med}[Z_b | s + b] = s / \sqrt{b}$. If $s \ll b$ does not hold, a better approximation is

$$\text{med}[Z_b | s + b] = \sqrt{2 \left((s + b) \ln \left(1 + \frac{s}{b} \right) - s \right)} . \quad (10)$$

Using Eq. (10), find me median significance for $x_{\text{cut}} = 0.1$. If you have time, try to write a program to find the value of x_{cut} that maximizes the median significance.

1(g) Now suppose that for each event we do not simply count the events having x in a certain region but we design a test that exploits each measured value in the entire range $0 \leq x \leq 1$. Thus there is no cut on x and in here we use $s = 10$ and $b = 100$ to refer to the total expected numbers of signal and background events. The data consist of the number n of events, which follows a Poisson distribution with mean of $s + b$, and the n values $x_1, \dots, x_n$.

The Poisson probability to find n events can be written in terms of a strength parameter μ as

$$P(n|\mu) = \frac{(\mu s + b)^n}{n!} e^{-(\mu s + b)} \quad (11)$$

where $\mu = 0$ corresponds to the background only hypothesis and $\mu = 1$ adds to this the contribution from the signal. The joint distribution for $\mathbf{x} = (x_1, \dots, x_n)$ given n is

$$f(\mathbf{x}|n, \mu) = \prod_{i=1}^n \left[\frac{\mu s}{\mu s + b} f(x_i|s) + \frac{b}{\mu s + b} f(x_i|b) \right] , \quad (12)$$

and the full likelihood is therefore

$$\begin{aligned}
L(\mu) &= P(n, \mathbf{x}|\mu) = P(n|\mu)f(\mathbf{x}|n, \mu) \\
&= \frac{(\mu s + b)^n}{n!} e^{-(\mu s + b)} \prod_{i=1}^n \left[\frac{\mu s}{\mu s + b} f(x_i|s) + \frac{b}{\mu s + b} f(x_i|b) \right] .
\end{aligned} \tag{13}$$

We can define a statistic q to test the background-only hypothesis as as

$$q = \sum_{i=1}^n \ln \left[1 + \frac{s}{b} \frac{f(x_i|s)}{f(x_i|b)} \right] = \ln \frac{L(1)}{L(0)} + C , \tag{14}$$

where C is a constant that can be dropped. This is a monotonic function of the likelihood ratio $L(1)/L(0)$ and thus according to the Neyman-Pearson lemma gives a test of $\mu = 0$ with the highest possible sensitivity (highest power with respect to the alternative of $\mu = 1$).

Run the program `hypTestMC.py`. This will produce histograms of q under the b and $s + b$ hypotheses, corresponding to the distributions $f(q|\mu)$ for $\mu = 0$ and $\mu = 1$, respectively. The program also finds the median q for the $s + b$ hypothesis, $\text{med}[q|s + b]$.

You should add code that finds the median p -value of the b -only hypothesis for the data generated under assumption of the $s + b$ hypothesis. That is, find the fraction of experiments simulated according to b -only that have $q > \text{med}[q|s + b]$. From this, find the p -value of the background-only hypothesis and the corresponding median significance $\text{med}[Z_b|s + b]$ (the sensitivity). Compare to the values you found from Eq. (10).

To find the value with reasonable accuracy you will need to simulate 10^7 experiments, which could take about 30 minutes to compute (the default value of `numExp` is set to 10^6). To test your code, use at first a smaller number of experiments.